Università degli Studi di Padova

Dipartimento Matematica
Dipartimento di Matematica "Tullio Levi-Civita"

Business, Economic and Financial Data Final Project

A Time Series Analysis of Cinema Admissions in Italy

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Academic Year:
2024/2025

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1 Introduction

Over the past few years, we have grown accustomed to watching movies on OTT (Over-The-Top) streaming—services that deliver content directly to viewers over the internet. These platforms have become an integral part of our daily routines, allowing us to enjoy movies in the comfort of our homes. However, this shift has made us forget the unique joy of sharing emotions with a crowd in a movie theater.

Particularly after the COVID-19 pandemic, it seems like the streaming platforms became dominant, and the overwhelming volume of content on these services has made it increasingly difficult for individual films to draw attention in cinemas. This led me to wonder: is traditional cinema culture approaching its end?

To explore further, this study focuses on the time series analysis and forecasting of cinema admissions in Italy.

1.1 Problem Definition

This study aims to address the following questions:

- Is traditional cinema culture coming to an end?
- Will OTT services dominate the distribution of movies in the future?
- Have the COVID-19 pandemic and/or OTT platforms impacted theater admissions significantly?

1.2 The Data Sources

The dataset used in this study consists of monthly data spanning from January 2006 to December 2023, complemented by a simplified annual dataset that includes only OTT Subscriptions and OTT Spending.

The data sources are as follows:

- **Società Italiana degli Autori ed Editori (SIAE):**
  Number of screenings in theaters, cinema admissions, ticket prices, audience spending [1].
- **International Video Federation (IVF) & Statista:**
  OTT subscriptions and OTT spending (annual data only) [4, 6].
- **Istituto Nazionale di Statistica (ISTAT):**
  Consumer Price Index (CPI) (Month-over-Month) for all items and CPI (MoM) for recreation and culture [3].
- **Istituto Superiore di Sanità (ISS):**
  Number of reported COVID-19 cases [2].
- **Google Trends:**
  Search interest for OTT platforms, used as a proxy for streaming consumption behavior [8].

2 Data Exploration & Preprocessing

2.1 Dataset Overview

This study uses two datasets: a monthly dataset (January 2006 - December 2023) and an annual dataset (2006 - 2023). These datasets provide detailed insights into cinema admissions, audience spending, economic indicators, and trends related to online streaming services (Netflix, Amazon Prime Video, etc.).

2.1.1 Monthly Dataset

The monthly dataset serves as the primary source for time series modeling. The key variables included are:

- **Cinema Admissions:** The number of individuals attending cinemas each month.
- **Number of Screenings:** The total number of movie screenings conducted per month.

- **Cinema Audience Spending:** The total expenditure by cinema-goers, including ticket sales and other expenses (in euros).
- **Average Spending Per Person:** The average amount spent per visitor in a given month (in euros).
- **Google Trends OTT:** A measure of online search interest in OTT (Over-the-Top) platforms. According to Statista's study on video-on-demand platforms, the most popular SVOD services in Italy as of June 2023 was Netflix, second came Amazon Prime Video, and Disney+ followed behind. [7] Therefore Google Trends for these platforms serve as an estimate of consumer engagement with streaming services.
- **CPI (All Items):** The month-over-month Consumer Price Index for all goods and services, reflecting general inflation.
- **CPI (Culture):** The CPI specific to cultural and entertainment-related expenses.
- **COVID-19 Cases:** The number of reported COVID-19 cases per month.

2.1.2 Annual Dataset

The annual dataset provides OTT Subscriptions, OTT Spending in annual format and aggregated yearly data on cinema attendance. This dataset is used to examine long-term competition in cinema admissions and OTT subscriptions. The key variables included in the annual dataset are:

- **OTT Subscriptions:** The total number of active OTT platform subscriptions per year.
- **OTT Spending:** The total consumer expenditure on OTT services (in euros).
- **Cinema Admissions:** The yearly total of cinema admissions.

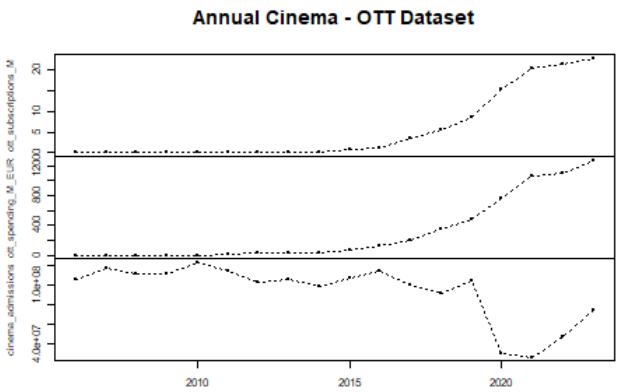

Figure 1: Overview of the Annual OTT Dataset

2.2 Handling Missing Values

The dataset initially lacked monthly data for cinema-related features (audience admissions, number of screenings, audience spending, average spending per person) for the year 2013, with only the total annual values available. To address this issue, monthly values were imputed using a proportion-based approach to ensure consistency with historical trends and seasonality. To impute the missing values, monthly averages from the adjacent years (2011, 2012, 2014, and 2015) were used to calculate proportional distributions. These proportions were then applied to the total annual values for 2013.

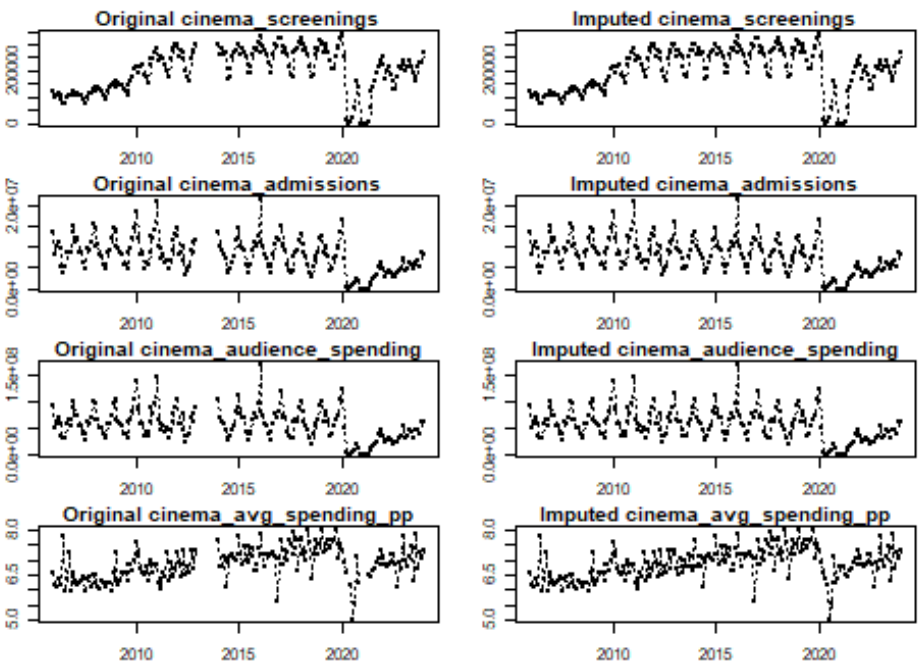

Figure 2: Original vs. Imputed Monthly Dataset

Additionally, a small number of missing values in 2020 and 2021 (likely due to disrupted operations during the COVID-19 period lock-downs) were replaced with the average ticket price for those respective years.

The annual dataset doesn't have missing values.

2.3 Collinearity Check

Before proceeding with modeling, it is important to examine potential collinearity issues among the predictor variables.

2.3.1 Correlation Analysis for Monthly Data

To assess the degree of linear relationships between variables, a correlation matrix was computed for the monthly dataset.

- **Cinema Admissions and Audience Spending** exhibit a very strong positive correlation.
- **Cinema Admissions and Number of Screenings** are also positively correlated.
- **Google Trends OTT and Cinema Admissions** display a moderate negative correlation.
- Inflation Indicators (CPI All Items, CPI Culture) show weak correlations with other variables.

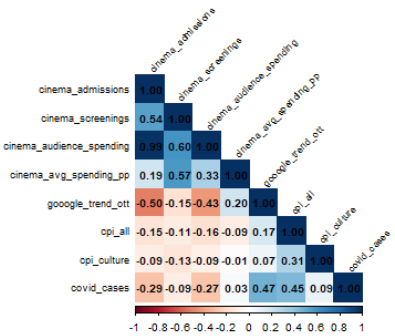

Figure 3: Correlation Matrix for Monthly Data

2.3.2 Variance Inflation Factor (VIF) Analysis for Monthly Data

While the correlation matrix provides valuable insights, it does not explicitly indicate whether collinearity poses a problem for regression models. Therefore, VIF analysis was conducted.

All VIF values are below 5, suggesting that multicollinearity does not pose a major issue for regression modeling. Cinema Screenings has the highest VIF value (2.13), followed by Cinema Audience Spending (1.98), but both remain well within the acceptable range.

Variable	VIF Value
Cinema Screenings	2.13
Cinema Audience Spending	1.98
Cinema Avg Spending Per Person	1.77
OTT Google Trends	1.75
CPI (All Items)	1.39
CPI (Culture)	1.13
COVID-19 Cases	1.58

Table 1: Variance Inflation Factor (VIF) for Monthly Dataset

2.3.3 OTT Google Trends as a Proxy for OTT Subscriptions and Spending

Since OTT Subscriptions and OTT Spending are only available in annual format, this section examines whether Google Trends OTT data can serve as a proxy for monthly modeling.

The correlation matrix presented in Table 2 confirms a strong relationship between the variables.

Variable Pair	Correlation Coeff.
OTT Subscriptions vs. OTT Spending	0.9981
OTT Subscriptions vs. Google Trends OTT	0.9379
OTT Spending vs. Google Trends OTT	0.9344

Table 2: Correlation Between Google Trends OTT and OTT Data

The results strongly suggest that Google Trends OTT search activity closely follows OTT Subscriptions and Spending trends over time. This supports the assumption that Google Trends data can serve as a reliable proxy for modeling OTT behavior. The visualization suggest that search interest in OTT platforms generally moves in sync with actual market adoption, though there may be a lagged relationship where increases in search activity trigger corresponding rises in subscriptions and spending.

.5Based on the high correlation values and trend alignment, Google Trends OTT data can be considered a viable proxy for monthly OTT Subscriptions and Spending.

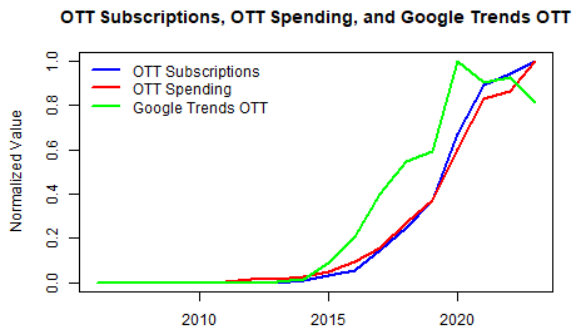

Figure 4: Comparison of Annual OTT and Google Trends OTT (Normalized)

2.4 Time Series Decomposition

Before testing models on the data, an in-depth analysis of the Cinema Admissions time series is performed to examine trend, seasonality, and any possible disruptions.

2.4.1 Time Series and Autocorrelation Analysis

To have an understanding of the data, the raw time series and its correlograms are plotted in Figure 5.

- The time series exhibits strong seasonality
- A downward trend is visible after 2020, indicating a structural break caused by the COVID-19 pandemic.
- The correlogram shows significant autocorrelations at seasonal lags, confirming the presence of seasonality.

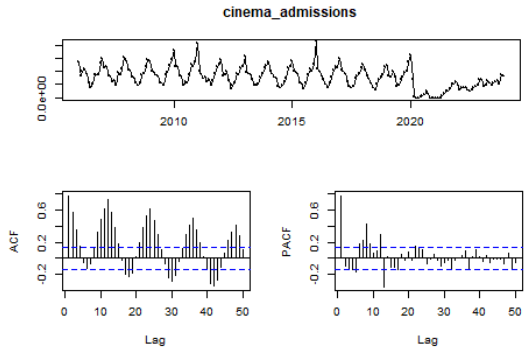

Figure 5: Time Series and Autocorrelation Analysis for Cinema Admissions

2.4.2 Seasonal Pattern Analysis

Cinema admissions data reveals distinct seasonal behavior, visualized in Figure 6.

- Admissions exhibit a recurring seasonal cycle, with peaks in January and December (likely due to holiday season).
- Lowest admissions are observed during summer season.
- The years 2020 and 2021 deviate significantly from seasonal behavior, indicating the COVID-19 impact.
- The recovery in 2022 and 2023 suggests a recovery, but levels remain lower than earlier years.

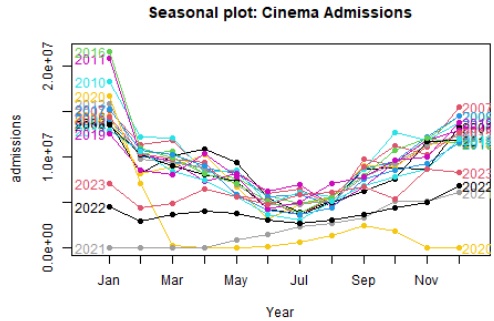

Figure 6: Seasonal Plot: Cinema Admissions (Yearly Comparison)

2.4.3 Decomposing the Time Series

The Seasonal-Trend decomposition using LOESS (STL) method is used to break down the time series into trend, seasonal, and residual components. It provides a useful way for capturing changing seasonality.

- A stable trend until 2019, followed by a sharp decline in 2020 and partial recovery in 2022-2023.
- The seasonal component remains stable across the entire period, reinforcing the consistency of seasonal patterns.
- Large residual fluctuations during the Covid-19 period, indicating high unpredictability.

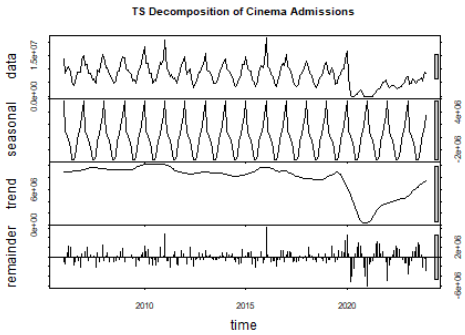

Figure 7: Decomposing the Time Series of Cinema Admissions

3 Time Series Modeling

Time series modeling allows us to analyze and predict future values of a sequence of data points, ordered in time. For this project, various methods and models were implemented and assessed based on their predictive power and interpretability. Below, the results and implications derived from the models can be found.

3.1 Multiple Linear Regression with Time Series

3.1.1 TSLM wih All Predictors

To understand the factors influencing monthly cinema admissions in Italy, a multiple linear regression model was fitted using all explanatory variables, including trend and seasonality.

The model shows **a strong fit (R^2 = 0.9954)**, indicating that nearly all variation in admissions is explained by the included variables. **Cinema spending** and **average spending per person** were highly significant, confirming their strong influence. Seasonal dummies(especially for winter months) were also significant, highlighting **consistent peaks in December and**

January. The **negative and significant trend** coefficient supports the hypothesis of a gradual decline in traditional cinema attendance.

Variables like Google Trends OTT, cinema screenings, and COVID-19 cases were not significant in this full model, though they may have relevance in other specifications.

However, **residual diagnostics revealed autocorrelation (Durbin-Watson = 1.0635)**, which limits the reliability of standard errors. This suggests the need for alternative models that better account for temporal dependencies.

3.1.2 Stepwise Linear Regression

To improve the model's efficiency, a stepwise regression was performed using both forward and backward selection based on the AIC criterion.

The final model retained key predictors: **cinema screenings, cinema spending, average spending per person, CPI (all items), trend, and seasonal dummies**. Variables like COVID-19 cases, Google Trends OTT, and CPI (culture) were excluded, indicating their limited contribution once other factors are considered.

This refined model maintains a high R-squared value with a slightly improved AIC score, comparable to the full model, suggesting better generalization. The **trend coefficient remained negative and significant**, again supporting the view that cinema attendance is declining over time.

Seasonality remained a strong component, with peaks in winter months still visible in the predictions. Visual comparison in Figure 9 confirms that the model captures key patterns in the data well.

Despite the model's parsimony and performance, **autocorrelation persists in the residuals (Durbin-Watson = 1.0635)**. This reinforces the need to explore other models that can better accommodate temporal dependencies in the data.

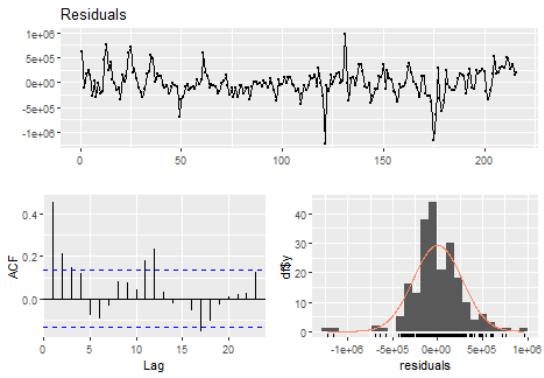

Figure 8: Residuals of Stepwise TSLM

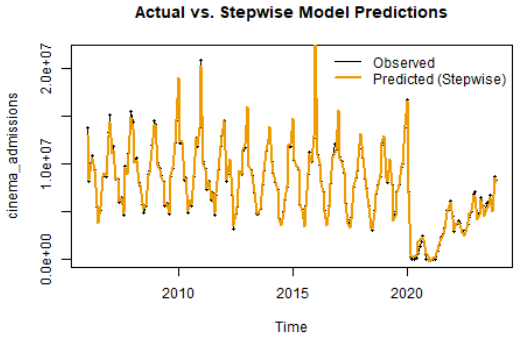

Figure 9: Fitted Stepwise TSLM Model

3.2 ARIMA Models

To model the time-dependent nature of cinema admissions, ARIMA models were applied after differencing the series to achieve stationarity. Both first-order and seasonal differencing were required to address the underlying trend and strong seasonality in the data.

3.2.1 ARIMA

Among the tested configurations, **ARIMA(2,1,5)(0,1,1)[12]** was selected as a strong baseline model.

This specification achieved an **AIC of 6422.8**, outperforming simpler alternatives like auto.ARIMA and effectively capturing the seasonal dynamics as seen in Figure 10. Residual diagnostics confirmed model adequacy, with the Ljung-Box test (p = 0.1602) indicating **no significant autocorrelation**.

While not the best-performing model overall, this ARIMA setup represents a considerable improvement over linear models in handling autocorrelation and serves as a reliable foundation for building more advanced models with external predictors.

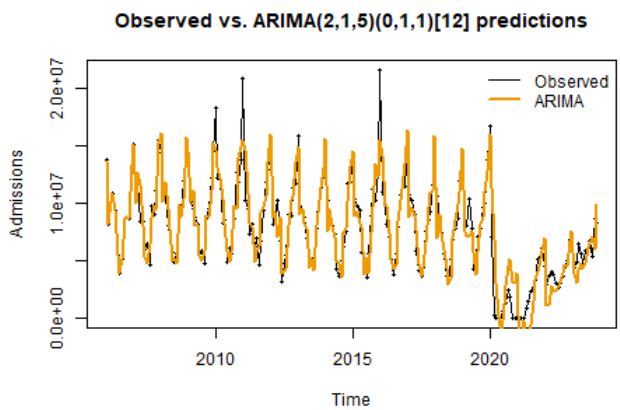

Figure 10: Fitted ARIMA Model

3.2.2 ARIMAx

To improve predictive accuracy, an ARIMAx model was implemented by incorporating external regressors, i.e. **cinema screenings, cinema spending, average spending per person, and CPI (all items)**, alongside the autoregressive and seasonal components. These variables were selected based on the results of the stepwise linear regression, which identified them as the most significant predictors.

Several ARIMAx configurations were tested, and **ARIMAx(1,1,2)(0,1,1)[12]** achieved the best overall performance. It recorded **the lowest AIC (5632.803)** and the highest R^2 (0.997) among all models considered, indicating strong explanatory power and model fit.

Residual diagnostics confirmed the model's adequacy, with **no significant autocorrelation remaining (Ljung-Box p = 0.7824)** as shown in Figure 11. The predicted values aligned closely with the observed data, accurately reflecting seasonal peaks and long-term trends as seen from Figure 12.

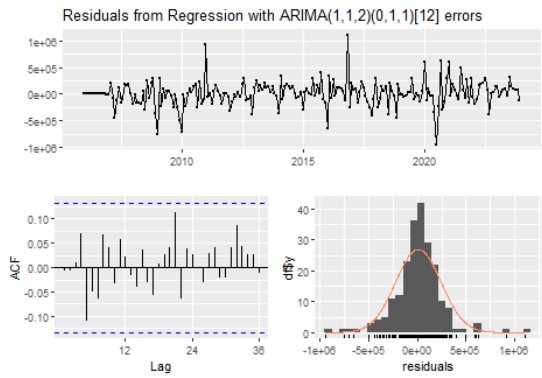

Figure 11: Residuals of ARIMAx Model

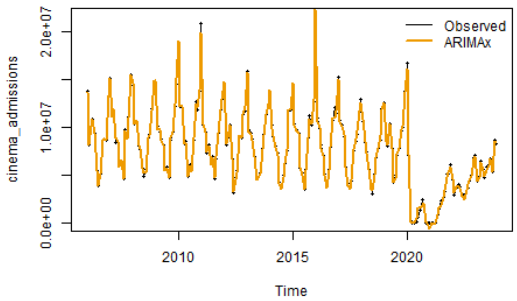

Figure 12: Fitted ARIMAx Model

3.3 Diffusion Models

Diffusion models describe how products, technologies, or behaviors spread through a population over time. In this context, they are used to model the adoption of cinema-going behavior. The models are based on three key parameters: the **innovation coefficient (p)** representing spontaneous adoption, the **imitation coefficient (q)** capturing influence from prior adopters, and the **market potential (m)** indicating the maximum adoption level. These models provide a complementary perspective to traditional time series methods for long-term trend analysis.

3.3.1 Bass Model

The Bass Model was applied **to estimate baseline diffusion parameters (p, q, m)** for use in other diffusion models. While it achieved a perfect R^2 (1.0) on cumulative data, this is misleading as the model does not account for seasonality and performs poorly on short-term dynamics. Residuals showed strong autocorrelation, with an extremely low Ljung-Box p-value (< 2.2e-16), confirming its limitations. The model serves as a preliminary step rather than a viable forecasting tool.

3.3.2 Generalized Bass Model (GBM)

The GBM extends the standard Bass framework by incorporating external shocks to capture abrupt disruptions in adoption behavior. In this case, **two exponential shocks** were added to represent the impact of COVID-19 lockdowns in Italy, corresponding to **March 2020** and **November 2020**.

These dates reflect key government actions: on 11 March 2020, Italy implemented a nationwide lockdown, shutting down non-essential services and restricting movement. After a gradual reopening phase, a second wave of restrictions began in November 2020, introducing region-based lockdowns due to rising infections and healthcare pressure. [5]

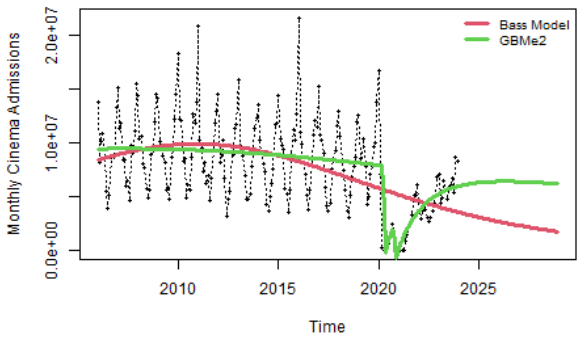

Figure 13: Bass Model and GBM with Exponential Shocks

Residual analysis revealed strong autocorrelation (Ljung-Box p < 2.2e-16), indicating the model is insufficient for detailed forecasting.

Nonetheless, GBM successfully captured the trend behavior, including the sharp drop and partial recovery caused by the pandemic. As shown in the Figure 13, the Bass Model forecasts a more pessimistic decline in cinema attendance, while GBM, accounting for the COVID shocks, projects a more optimistic recovery trajectory. However, both models suggest an overall **negative trend in cinema admissions**. The estimated GBM trend will now be refined and integrated with ARIMA in the next section to better capture seasonality.

3.3.3 GBM with SARIMAx Refinement

To improve the forecasting ability of GBM, a hybrid GBM + SARIMAx model was implemented. This approach combines the long-term trend captured by the GBM (including pandemic shocks) with an ARIMA component that handles short-term fluctuations and seasonality.

An ARIMA(2,1,0)(0,1,2)[12] model was fitted to cumulative cinema admissions, using the GBM's fitted values as an external regressor as seen in Figure 14. The model achieved an AIC of 6379.206 and reduced residual autocorrelation, but the Ljung-Box p-value of 0.044 indicates that **autocorrelation is still slightly significant** and has not been fully eliminated. The R^2 value of 0.718, while lower than other models, reflects a more realistic fit to instantaneous, non-cumulative behavior.

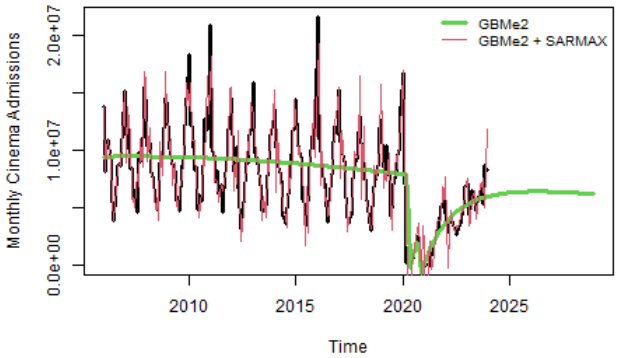

Figure 14: GBM and GBM+SARIMAx Models

Compared to the best-performing ARIMAx model, which achieved near-perfect in-sample metrics (R^2 = 0.997, AIC = 5632.8), the hybrid model sacrifices some in-sample fit but potentially avoids overfitting. By combining a structurally interpretable diffusion trend with a SARIMAx refinement, this approach may offer better generalization to future conditions, especially in contexts marked by structural disruptions and evolving dynamics.

3.3.4 Unbalanced Competition Regime Change Diachronic Model (UCRCD)

This section investigates the evolving competition between cinema and OTT platforms using the UCRCD model, which captures how two diffusion processes interact over time.

Unlike previous models, this section is based on the annual dataset, as it aligns with the available OTT subscription data. To estimate initial parameters, diffusion models (Bass, GBM, GGM) were first fitted independently to both series.

Due to the differing scales of the two series, normalized values were used to ensure numerical stability and allow focus on relative market dynamics. Although normalization is not always standard, it is a practical choice here since the goal is to explore market interaction patterns, not absolute values.

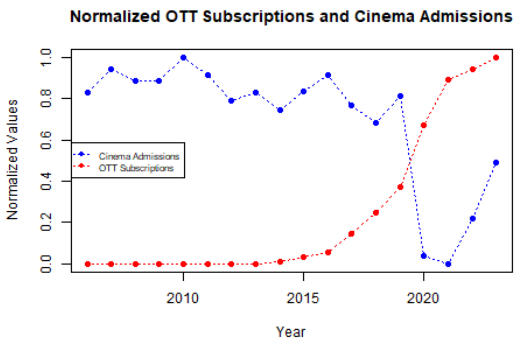

Figure 15: OTT Subscriptions and Cinema Admissions (Normalized)

The UCRCD model then jointly modeled the diffusion patterns of both OTT subscriptions and cinema admissions, allowing for the estimation of cross-market influences. Results indicate a clear **asymmetric competitive relationship**:

- The cross-imitation coefficient from OTT to cinema, q_{1c} = -0.25893, is significantly negative (p = 0.0209). This implies that **increases in OTT subscriptions are strongly associated with decreases in cinema admissions**. In behavioral terms, this suggests that as OTT services gained popularity and accessibility, consumer willingness to attend the cinema declined in response.
- On the other hand, the cross-imitation from cinema to OTT, calculated as $q_2 - \gamma$ = 1.110295 - 1.109853 = 0.000442, is **extremely small and not significant**, indicating that cinema activity had virtually **no impact on OTT subscription** dynamics. This lack of influence confirms that the OTT market grew independently of cinema trends, likely driven more by technological adoption and pandemic-related behavioral shifts than by traditional media consumption patterns.

These findings point to a **one-sided competitive displacement**, where OTT services increasingly exhaust cinema market share, but not vice versa. The dynamic became particularly evident after the COVID-19 period, which acted as a regime-change catalyst—accelerating OTT adoption while heavily disrupting traditional cinema habits.

Conclusion: OTT services have gained dominance in market dynamics, while cinema has lost its competitive influence in shaping consumer adoption.

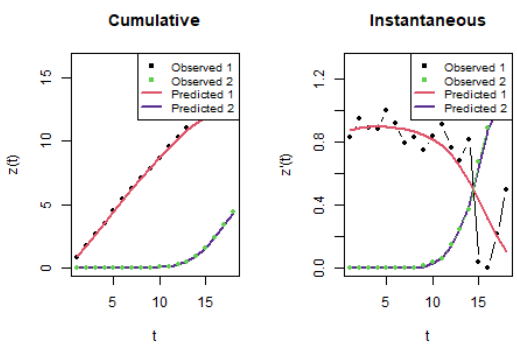

Figure 16: Fitted UCRCD model

3.4 Prophet Model

Prophet is a relatively recent forecasting tool developed by Facebook, designed to handle time series data with strong seasonal effects and irregular events. It is based on a **Generalized Additive Model (GAM)**, where the time series is decomposed into additive (or multiplicative) components: trend, seasonality, and holiday effects (Figure 17). Each of these components is modeled using flexible smoothing functions (such as piecewise linear or logistic growth for trend, Fourier series for seasonality, and user-defined events for holidays), which makes the model capable of capturing complex, non-linear behaviors in the data.

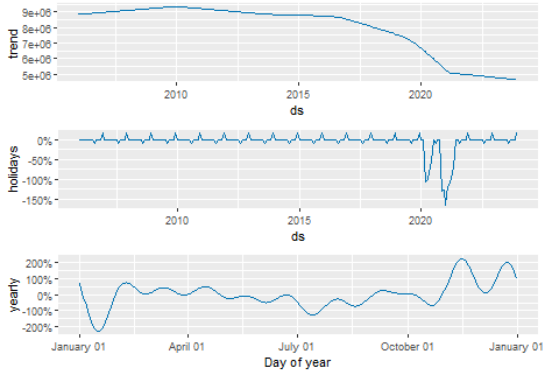

Figure 17: Decomposed Time Series Components

One of Prophet's key strengths is its **interpretability: the contribution of each component can be visualized separately**, offering insights into both long-term patterns and specific external events that influence the time series.

In this analysis, Prophet was applied with multiplicative seasonality and custom-defined holidays to account for known irregularities in the cinema market. Notably, lockdown periods due to COVID-19 in March and November 2020 were encoded as **special holidays**. Additional yearly events such as Christmas and Ferragosto were also included to reflect typical seasonal variations.

The model effectively captured the long-term declining trend and sharp drops in cinema attendance during lockdowns. However, residual diagnostics revealed **significant autocorrelation** (Ljung-Box p-value = 1.18e-12), indicating that the model does not fully account for temporal dependencies as shown in Figure 18.

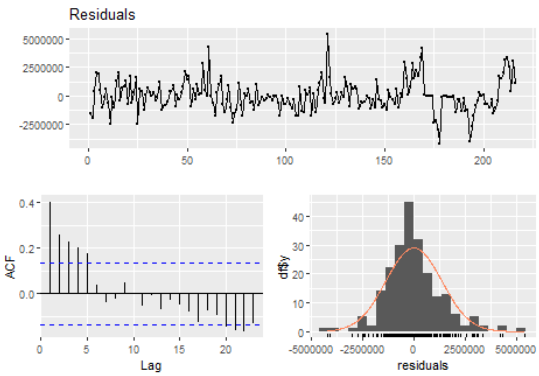

Figure 18: Residuals of Prophet Model

3.4.1 Prophet with SARMAx Refinement

To address the limitations of the base Prophet model, particularly the presence of significant autocorrelation in its residuals, a SARMAx refinement was applied. In this hybrid approach, the fitted **Prophet model was used as external regressor** in an ARIMA model that incorporates both autoregressive and seasonal components.

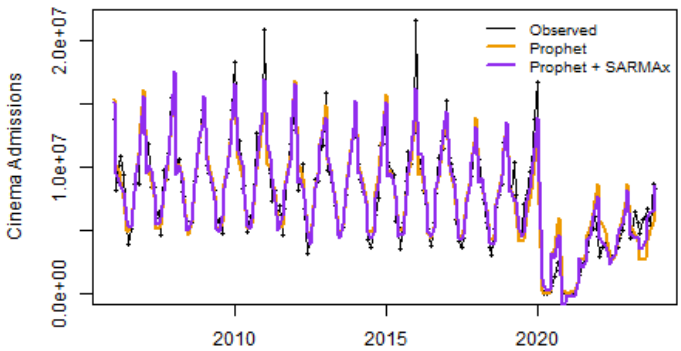

Figure 19: Fitted Prophet and Prophet+SARMAx Models

The resulting model, ARIMA(1,0,1)(2,0,0)[12], provided a stronger overall fit compared to the base Prophet model (Figure 19). The residuals showed **no significant autocorrelation** (Ljung-Box p-value = 0.5968) as shown in Figure 20, confirming a substantial improvement in error structure. This correction directly addresses one of the key shortcomings of the original Prophet model. Additionally, the hybrid model improved upon Prophet's forecasting performance in terms of both MAE and RMSE, while retaining the interpretability and trend decomposition strengths of the original model.

Although the R^2 = 0.907 of this hybrid model does not surpass that of the best ARIMAx model (R^2 = 0.997), the difference highlights an important trade-off. The Prophet + SARMAx model offers a more regularized structure that combines Prophet's flexibility in handling trend shifts, seasonality, and external shocks (such as holidays and lockdowns) with ARIMA's strength in modeling autocorrelation. This makes the Prophet + SARMAx approach **a strong candidate for forecasting**, where generalizability and robustness often matter more than maximizing in-sample fit.

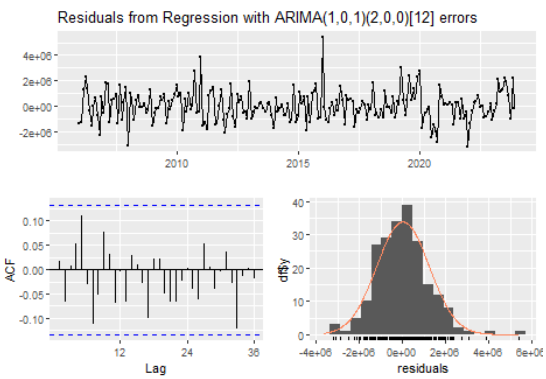

Figure 20: Residuals of Prophet+SARMAx Model

3.5 Exponential Smoothing

3.5.1 Holt-Winters method

Exponential smoothing models forecast time series data by giving more weight to recent observations. The Holt-Winters method extends this by including both trend and seasonality, making it suitable for structured seasonal data.

The Holt-Winters additive method was applied to model trend and seasonality in cinema admissions. The model produced forecasts using a high level smoothing parameter ($\alpha = 0.7404$), but extremely low values for trend and seasonal smoothing (β and $\gamma = 0.0001$), suggesting limited responsiveness to changes in those components.

Despite capturing the overall structure, the model performed poorly compared to other approaches. It showed **strong residual autocorrelation** (Ljung-Box p-value = 2.31e-05), and had the **highest AIC (7384.40)**, indicating weak predictive performance.

4 Forecasting

4.1 Model Performances

The performance of all models was evaluated using statistical metrics such as AIC, R^2, Ljung-Box p-value, MAE, and RMSE. Due to the presence of a structural break in recent years caused by the COVID-19 pandemic, standard train-test or cross-validation procedures may produce misleading error metrics. Specifically, the forecasting performance of models trained before COVID and tested on post-COVID data is affected by substantial non-stationarity.

Therefore, in this project, models were evaluated using in-sample metrics and overall fit to the full time series. The best-performing models were selected based on their ability to capture both trend and seasonality, while controlling for residual autocorrelation. The highest-performing, acceptable models are highlighted in the table below:

Model	AIC	DW	Ljung-Box p-val	R^2	MAE	RMSE
TSLM - Full	6054.269	1.065	NA	0.995	193,067.61	267,895.01
TSLM - Stepwise	6050.035	1.064	NA	0.995	195,166.90	268,992.46
ARIMA(2,1,5)(0,1,1)[12]	6422.800	NA	0.160	0.833	1,171,316.97	1,621,661.60
auto.ARIMA(1,0,2)(0,1,1)[12]	6448.711	NA	0.108	0.828	1,158,030.22	1,647,445.01
auto.ARIMAx(3,1,0)(2,0,1)[12]	5974.656	NA	0.047	0.996	170,544.06	245,186.45
ARIMAx(1,1,2)(0,1,1)[12]	5632.803	0.782	0.997	0.997	139,146.56	233,099.99
Bass Model	NA	NA	< 2.2e-16	1.000	2,772,250.27	3,480,549.43
GBMe2	NA	NA	< 2.2e-16	1.000	2,307,165.36	3,106,753.77
GBMe2 + ARIMAx	6379.206	NA	0.044	0.718	1,061,865.94	1,490,790.20
Prophet	NA	NA	1.181e-12	0.885	975,508.94	1,348,348.68
Prophet + SARMAx	6676.570	0.597	0.907	0.907	929,898.93	1,211,338.20
Holt-Winters Exponential	7384.404	NA	2.311e-05	0.824	1,248,622.75	1,668,082.59

Table 3: Comparison of model performance metrics with highlighted acceptable models

Among the tested models, the **ARIMAx(1,1,2)(0,1,1)[12] provided the best in-sample performance** with the lowest MAE and RMSE, as well as a high R^2 value, indicating an almost perfect fit to the observed data. However, the high complexity of this model and its near-perfect in-sample fit raise concerns about potential overfitting.

The GBMe2 + ARIMAx hybrid model, while conceptually promising due to its incorporation of trend and shock information from GBM, exhibited residual autocorrelation, suggesting that some systematic patterns remain unexplained. Despite its moderate predictive performance, this limitation reduces its reliability for forecasting and it will not be selected.

In contrast, **Prophet + SARMAx demonstrated both strong performance and well-behaved residuals** ($p = 0.907$), making it a robust and interpretable choice for forecasting. Additionally, although the best-performing ARIMAx model has a risk of overfitting, it can still serve as a benchmark for comparison in future projections due to its superior fit and minimal residual autocorrelation.

Based on the overall assessment, two models are selected for future forecasting of cinema admissions:

- Prophet + SARMAx
- ARIMAx(1,1,2)(0,1,1)[12]

4.2 Forecast with Prophet + SARMAx

Future values were produced by extending the Prophet model and refining its residuals through ARIMA to capture remaining autocorrelation.

The forecasted trajectory in Figure 21 (shown in blue) suggests **a continuation of the long-term declining trend** in cinema admissions. Although a partial recovery is visible after the COVID-19 shock, the model does not anticipate a return to pre-pandemic levels. This indicates that the recovery may have plateaued and **a structural shift in cinema attendance behavior could be persisting**.

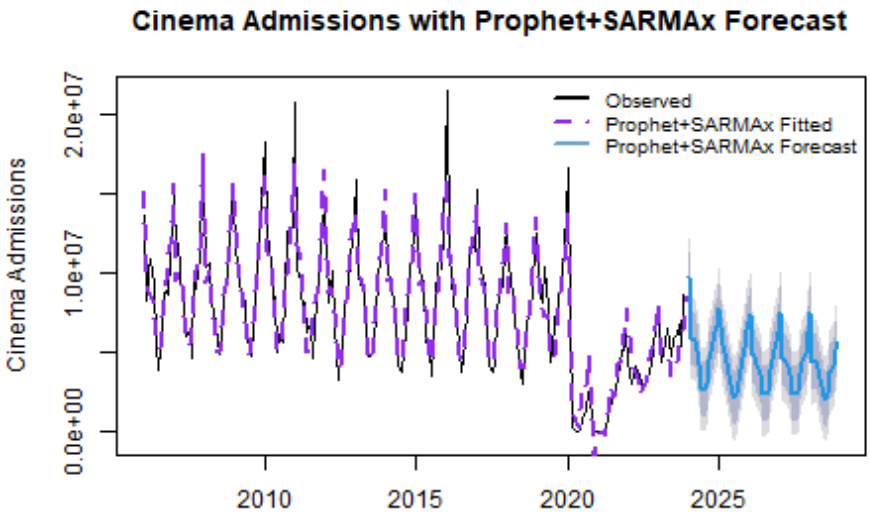

Figure 21: Cinema Admissions with Prophet+SARMAx Forecast

4.3 Forecast with ARIMAx

The final forecasting approach applies the best-performing ARIMAx(1,1,2)(0,1,1)[12] which was selected based on its superior in-sample performance. This model integrates several **external regressors—screening counts, cinema spending, average spending per person, and the consumer price index (CPI)**—each of which was forecasted individually using ARIMA models. The resulting predictions were combir

Despite the model's strong fit to the historical data, its tendency to overfit was noted in the performance evaluation. Nonetheless, this forecast was carried out for experimental purposes, acknowledging the **overfitting risk** and with the aim of exploring the model's potential for future scenarios.

The ARIMAx forecast projects a clear **upward trend** as seen from Figure 22 in cinema admissions, suggesting a stronger post-COVID recovery. This contrasts with the more conservative forecast produced by the Prophet+SARMAx model, which extended the overall declining pattern observed in previous years. This comparison emphasizes how model structure and the choice of predictors can significantly influence long-term forecasts.

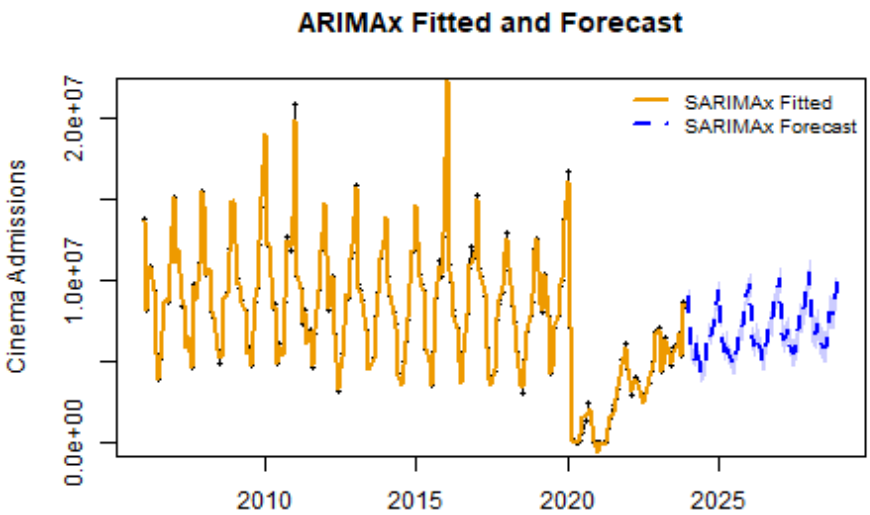

Figure 22: Cinema Admissions with ARIMAx Forecast

Interestingly, this more optimistic outlook brings a hint of hope for the traditional cinema experience—perhaps the decline of cinema culture is not as inevitable as recent trends have suggested.

5 Conclusion

This study set out to explore the dynamics of cinema admissions in Italy in light of structural changes caused by the rise of OTT platforms and the COVID-19 pandemic. Through a comprehensive set of time series and diffusion models the analysis aimed to understand both the past behavior and future prospects of cinema attendance.

The results consistently showed **a declining trend in cinema admissions**, particularly accelerated during and after the COVID-19 pandemic. While some recovery was observed in the post-COVID period, most models indicated that a full return to pre-pandemic levels is unlikely in the near future. This suggests **a structural shift in consumer behavior** rather than a temporary shock.

The UCRCD model offered further insight by quantifying the interaction between cinema and OTT platforms. It showed that **OTT growth negatively impacts cinema admissions**, while cinema has virtually no effect on OTT subscriptions. This asymmetry points to **a one-sided competitive displacement**, where OTT services increasingly dominate the entertainment market.

Yet, not all outlooks were pessimistic. The forecast from the best-performing ARIMAx model suggested a more optimistic recovery scenario, projecting **a gradual upward trend** in cinema attendance. While this forecast must be interpreted with caution due to potential overfitting, it highlights the possibility that cinema culture may not be fading entirely.

So, if we return to the questions posed at the beginning of this project:

- **Is traditional cinema culture coming to an end?**

  The evidence points to a significant decline, but not necessarily a definitive end. A potential for stabilization or partial recovery remains, especially under favorable conditions.

- **Will OTT services dominate the distribution of movies in the future?**

  The findings suggest that OTT platforms are increasingly dominant and structurally favored, making them likely to continue expanding their role in movie distribution.

- **Have the COVID-19 pandemic and/or OTT platforms impacted theater admissions significantly?**

  The analysis indicates that both factors have had a substantial impact. COVID-19 acted as a major disruption, while OTT platforms appear to support a longer-term shift in consumer preferences away from traditional cinema.

While the landscape of movie consumption is clearly changing, cinema may still find a place—perhaps more niche, event-based, or culturally embedded—in a future shaped by digital convenience and evolving viewer preferences.

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